

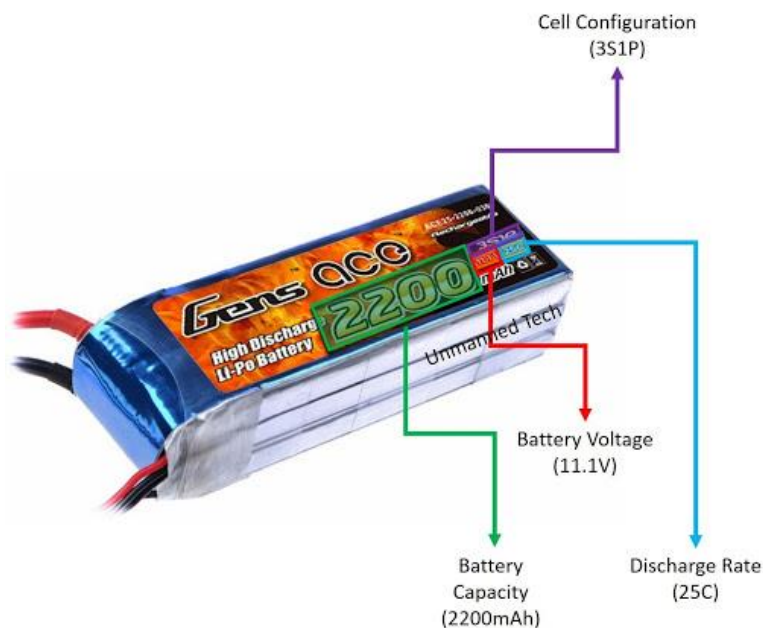
What's the purpose of this document?

1. What's Lipo battery, how we choose it, maintenance and safety
2. Why go for Gens Tattu
3. Power system in drones (Calculation and Setup – requirements and other equipment)
4. We can use it as living document – in creating wiring and power consumption calculation for each setup.
5. Things to remember (possible mistakes as a beginner)

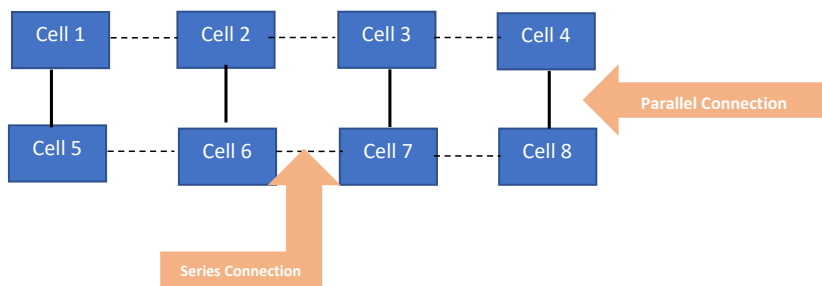
1. What's Lipo battery, how we choose it, maintenance and safety

<https://genstattu.com/bw/?srsltid=AfmBOopodfFAjaEjwRezK1F7FlVWKGwyjo6F0sx3ZC-WajeeGdMovp0b>

- Lipo (lithium Polymer) is a rechargeable battery of lithium-ion technology using a polymer electrolyte instead of a liquid one.
- High conductivity semisolid polymers form this electrolyte. They have higher capacity to hold charge in this form. When crystalized – mean they are not in good state anymore.
- LiPo battery have following parameters (Cell configuration, Capacity, Voltage and Discharge Rate)



- A battery is constructed from rectangular cells which are connected together to form the battery. A cell which can be considered a battery in itself, holds a nominal voltage of 3.7V. But when fully charged each cell max voltage is 4.2 V.
- The connection in series increase the voltage and the connection in parallel increase the capacity.
- For eg.; 2 cells in series battery hold 7.4V, 4 cells give 14.8 V, 6 cells 22.2 V
- So, we often see in battery that 4S2P, 6S1P which means number of cells in Series (S) and number of cells in parallel (P)
- Let's visualize 4S2P to gain insight about the voltage and capacity. Meaning 4 cells (4S) groups (row) of 2 (2P) connected in parallel.



- Series and parallel connection impact the voltage, capacity and discharge current. In series, voltage increase but capacity remains the same. In parallel connection, capacity increase and voltage remain the same.
- The series is important for power to motors. The latter can increase flight time. Voltage is the key for heavy-lift drones.
- Voltage: $4 * 3.7V = 14.8 V$
- Capacity is how much power (the amount of charge) a battery can hold.
- It is measured in **milliamp-hours (mAh)** or **amp-hours (Ah)** and determines how long a battery can power a device before needing a recharge.
- More capacity means more density of electrolyte in a cell which makes them heavier.
- Capacity (Ah) = $(\text{Current (A)} \times \text{Time (h)}) / \text{Discharge Rate}$
- For 5000 mAh = 5 Ah (1 mAh = 1000Ah) . Depends on the the discharge rate(c) (Amp)

More about Battery voltage and capacity:

- Voltage directly influences the RPM of the electric motor (brushless motors are rated by kV, which means 'RPM per Volt'). So if you have a brushless motor with a rating of 3,500kV, that motor will spin 3,500 RPM for every volt you apply to it. On a 2S LiPo battery, that motor will spin around 25,900 RPM. So the more voltage you have, the faster you're going to go. (Voltage impact motor, motor impact speed)
- Power (P) = Voltage (V) * Current (I)
- Capacity affects the speed, the more quick you can flying your plane, the less time your flight time is.

Discharge Rate(C) and Time

- It is known as C-Rating and denotes how fast the battery can deliver power safely. That is , maximum current from the battery without damage.
- Higher the C-Rating – more current output | Another thing to go for higher C-Rating is it can handle higher power loads. And have less voltage drops compared to low C-rating batteries.
- C -Rating is important as it prevents over heating and damage
- Max current (A) = C-Rating * Battery Capacity (Ah)
- For eg: 5000 mAh 30C Lipo battery can give continuous discharge current $\rightarrow 30C * 5Ah = 150 A$
- Here, Battery discharge calculation is important – to know flight time
- Discharge Time= BatteryCapacity(Ah) / Load Current(A)
- Given battery 1300mAh – i.e; 1.3Ah and we draw current at 2.6 A then the max flight time/ battery lasts 30 mins
- Higher the discharge current – faster the depletion of battery
- Battery discharge rate and time are inversely proportional – high discharge rate leads to short time.

Maintenance and Safety of battery

- Please read the charger instruction book before charging.
- Always check the voltage of Batteries before each charge session in order to ensure they are at or above the minimum safe starting voltage. If starting voltage is below recommended levels then Batteries have been over discharged or have experienced a failure and should NOT be charged.
- Always check the battery before charging for any type of damage. Check the battery packaging, wires and connectors for defects, which may cause a short circuit and eventual battery failure.
- Make sure you are using charge leads which are compatible with the battery connectors.
- User should check the polarity of the battery cable and charger lead carefully before the connection to avoid any short circuit.
- Always verify the charger is in good condition. A poor-quality charger can be dangerous.
- Be around while charging and make sure to monitor the charging process.
- Do not use a NIMH/ NICD/ LIFEPO4/ LEAD ACID charger. If the charger can support different battery types. Be absolutely sure to select the Lithium polymer (Lipo) mode on the charger. Failure to do so may cause a fire, which may result in personal injury and property damage.
- User should Always charge Batteries in an open area away from flammable materials, liquids and surfaces.
- Never charge Batteries inside of the model.
- Never charge batteries below freezing (0°C, 32°F)
- Never charge Batteries that are hot to the touch (above 100° F). DO NOT handle Batteries until they are cool.
- Always set the charger to the proper cell count and/or voltage listed on Batteries' labels.
- User should Always set the charger to the amp charge rate as listed on Batteries' labels.
- The charger should never be set to charge Batteries at a rate greater than 1C (One (1) times the capacity of Batteries in amp hours) unless another C rate is specified in the manufacturer's product documentation or the rate is preset as part of a specific battery and charger combination.
- DO NOT alter the charge rate once charging has begun.

- Never overcharge Batteries beyond the capacity listed on Batteries' labels.
- Never overcharge batteries over their rated maximal voltage (4.2V/cell for LiPo).
- It must choose series charging if two or more battery packs be connected in series for using.
- When charging, please make sure the surroundings is clean, without sundries around.
- Using lipo sack is suggested when charging to ensure the charging safety. Stop charging timely when the battery is fully charged.
- Do not connect the charger and the battery for a long time when the battery is fully charged.
- The recommended cut off voltage is 3.5V per cell. Never discharge a battery to level below 3V per cell.
- Never leave battery unattended during the discharging process.
- In case of emergency, discontinue the process immediately, disconnect the battery, place it in a safe area, and observe it for approximately one hour. This may cause the battery to leak and the reaction with air may cause the chemicals to ignite, resulting in a fire.
- A safe area should be outside of any building or vehicle and away from any combustible materials. A battery can still ignite even after one hour.
- check the conditions of the battery before use it or discharge it. Stop using it if user found the cells are not balanced or cells are puffing or weeping.
- Do not directly connect the terminals with metal objects. This will short-circuit Batteries, resulting in heat and electrical discharge.
- Never store loose Batteries together, the Batteries' terminals may contact one another causing a short circuit.
- Never storage Batteries to extreme temperatures or direct sunlight. The battery should be stored within -10°C ~ 45°C range environmental condition. If the battery has to be stored for a long time (Over 3 months), the environmental condition should be:
 - Temperature: 23±5°C
 - Humidity: 65±20%RH
 - The voltage for a long time storage should be 3.6V~3.9V range per cell
- Always disconnect Batteries when not in use and store Batteries in a non-conductive and fireproof container.

Choose battery

- Calculate how much total power to draw and plan the C-rating.
- Check the total weight of the vehicle (incl frame, motor, payload) and decide the max capacity. This is important for weight balance.
- Higher voltage means more cells – better power efficiency. Check the voltage setup needed for the equipment (motor, controller..)
- In short – high capacity means longer flight time, and high voltage means better efficiency.

2. Why go for Gens Tattu

- Premium Model Bau uses Tattu battery for their drones. Freely Alta X, tarot quadcopters uses tattu

Commented [SS1]: Gens supplier in Dusseldorf – Contacted, found Premium Modelbau is also a supplier – need help to contact them and enquire.

- Tattu has following reliability certificates – UN38.3 – for shipping, Material Safety Data sheet, RoHS and other ISO standards. Suitable for industrial, military, professional UAVs
- DJI Matrices for industrial use case have special modification and use Tattu
- Open FPV racing, DIY drone community using Tattu. A lot of videos, chats forums available for discussion and support.
- Holybro page suggested Tattu battery on page
- Reliability, stable voltage (no complaints on voltage sag) battery cycles. With people experience its understood tattu can last 300-500 battery cycles.

Commented [SS2]: if I remember correctly, DLR was using the same (visit from Cochstedt). Need Tushar's help to check

At this point, we don't consider using two batteries in a single drone. Multiple batteries need to be considered whether they are connected in series or parallel.

Commented [SS3]: This needs more analysis. For eg.,
1. Are we using two batteries for drone(motor), and payloads separately?
2.Or the secondary /backup
3.Weight management

<https://electronics.stackexchange.com/questions/681078/drone-power-distribution-board-for-multiple-battery-setup>

3. Power system in drones (Calculation and Setup – requirements and other equipment)

Battery recommendation by Holybro X650 - 6S 5000-12000mAh Lipo Battery with XT60

6S 10000 mAh Battery – Hovering – 30 mins flight time <https://holybro.com/collections/x650-kits/products/x650-kits>

Our choice:

1. 6000 mAh 22.2V 6S1P 30C
- Discharge current: $6Ah * 30C = 180A$
2. 12000 mAh 6S1P 22.2 V 30C
- Discharge current: 360A

Commented [SS4]: Paid attention to voltage. Need to consider weight before if we need 6S2p. Also, go for higher capacity or 6S2P – progressive learning from experimentation and experience

Commented [SS5]: We go for 30C better than 10-15C, weigh suitable for X650 and had more reviews than C-rating above for this particular drone weigh category

- Remember - low discharge current than needed – voltage drop, heating issues

- 22.2 V for 6S does not directly affect the discharge current but determines the power to motor

Power Consumption Calculation

- Estimate total power draw of all components
- We can create some table like

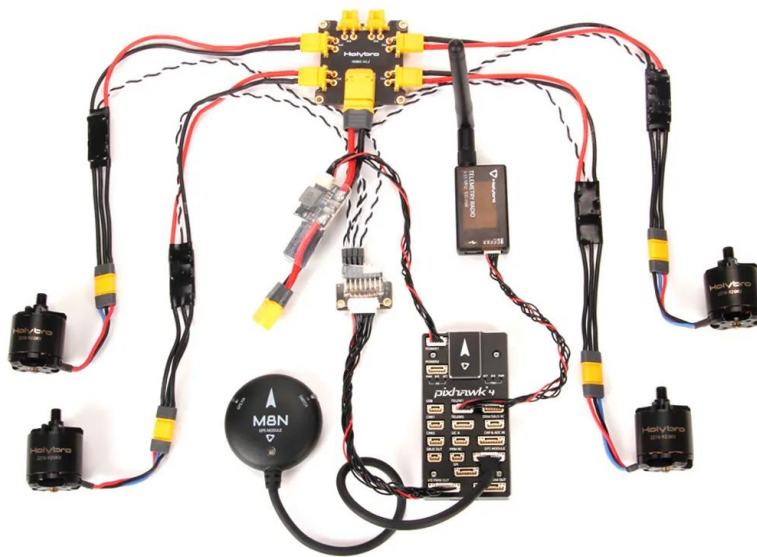
Component	Power (W)	Voltage(V)	Current (A)
Total Draw			

- Then we use Discharge time calculation to estimate flight time
Discharge Time = (Battery capacity (Ah) and Total Current Draw (A))
- This calculation can give us estimated flight time

Commented [SS6]: We can create this calculation for each sensor setup separately and validate in real world to gain experience. Then we can proceed to combine and fly. Because, powering with the socket /laptop on hardware-in-loop is different than using batteries.

The other components that help in power management – Power distribution board, Voltage regulators, Power modules, ESC, Power monitoring system

- Power distribution board (PDB) – Distributes the power from battery to ESC, flight controller and peripherals. It organizes the wiring (todo: yet to be explored).
- PDB have multiple power output – like 5V, 12V, etc..
- Used with Power module to support Pixhawk and telemetry unit
- https://holybro.com/products/power-distribution-board-pdb?pr_prod_strat=jac&pr_rec_id=d2718e667&pr_rec_pid=7192463737021&pr_ref_pid=7610651345085&pr_seq=uniform – Holybro pdb setup
-



- Some PDB comes with Battery elimination circuit (BEC) for voltage regulation - https://www.mybotshop.de/MAUCH-PL-BEC_2
- Products considered for PDB: Holybro PM07, PM02 power module, Matek PDB- XT60
- Voltage regulator steps down the voltage of 22.2V to 5V, 12V to prevent damage of flight controllers, camera and sensors.
- Two types of voltage regulators – linear and switching
- Linear regulators reduce the voltage by dissipating excessive power as heat (acts like variable resistor). Need heat sink and no EM interference, easy circuit.
- Switching regulator (buck convertor – step down voltage, boost convertor – step up voltage) turns on and off at high frequency (acts like inductor and capacitor energy storage circuit). Handles high power load. Can introduce EM interference.
- https://holybro.com/products/ubec-12a?srltid=AfmBOoqV1n6Wk7K17xhkrQmYpYEEyvNKH4bX4P4MltDW6Cagg_ShbbHK Holybro stepdown regulator

- We use digital power module for 6X Pixhawk - <https://holybro.com/products/pm02d-power-module?srltid=AfmBQooR7eyQ-YMeDnVz8CroJA7XY7aMq2S15rbQ5LqrllNeKj4N9Eph>
 - For camera we can consider - <https://www.premium-modellbau.de/matek-ubec-duo-4a-5v-12v-schaltbar-4a-5v-bec-fuer-2s-6s-lipo>
 - Other Matek options – Matek UBEC 12V 3A, Matek Mini BEC 5V 2A, Matek Dual BEC 5V/12V
 - ESCs – Electronic speed controller offered for Holybro X650 – Tekko32 65A , Holybro 45A ESC
 - Power module is to provide stable power to Pixhawk. There is analog and digital output module based on Pixhawk version. Pixhawk 6X for instance is only compatible with PM02D
 - Battery checker https://www.mybotshop.de/ISDT-BG-8S-Smart-Battery-Checker_2
1. https://docs.px4.io/main/en/flight_controller/pixhawk4.html#connectors
 2. https://docs.px4.io/main/en/assembly/quick_start_pixhawk4.html
 3. <https://cdn.sparkfun.com/assets/9/b/1/8/6/PM07-Quick-Start-Guide.pdf> (Power Mangt board)

Commented [SS7]: Todos: Check with camera, LiDAR, sensor compatibility

Commented [SS8]: To do: Need to study deep into motors and speed controllers
Why we need, individual relators to motors not sufficient?

Commented [SS9]: Todo: learn how to use for LiPo battery

4. We can use it as living document – in creating wiring and power consumption calculation for each setup.

2 things to check - Flight time based on power draw and wiring from battery to the peripherals – using 6000mAh 30C battery

Setup 1 – Holybro X650, Pixhawk, telemetry unit (433Mhz)

Components	Power (W)	Voltage(V)	Current(A)
Holybro Motor , ESC (Hovering)	Approx. 350W	22.2V	Approx. 15.8 A
Pixhawk (with GPS)	5W	Max 6V (4.75 – 5.25V)	1A
Telemetry device	Approx. 2W	5V	0.4A
Total Draw	357 W	Max 22.2V	17.2 A

This doesn't include any additional sensors. Therefore, 6000mAh 22.2V 30C 6S battery can provide more than required (180A). Flight time = 6Ah / 17.2 A = 0.348 hrs (time conversion - 0.348 *60 = 20.88 mins)

We will have approx 20 mins of flight

Setup 2 - with Intel RealSense D435 Camera

Components	Power (W)	Voltage(V)	Current(A)
Holybro Motor , ESC (Hovering)	Approx. 350W	22.2V	Approx. 15.8 A
Pixhawk (with GPS)	5W	Max 6V (4.75 – 5.25V)	1A
Telemetry device	Approx. 2W	5V	0.4A
Intel RealSense Camera D435	2.5W	5V	0.5A
Total Draw	359.5 W	Max 22.2V	17.7A

Flight time: 0.338 hrs – 20.33 mins

Close to 20 mins – Now the real question is how the power draw with the payload impact the flight time?

Commented [SS10]: Here is sample for small setup. I only used Power, voltage, Current as basic criteria. Kindly throw in ideas and suggestions
To do: learn about wiring for power and issues.

Commented [SS12]: According to some posts – 16,3A as 360W power draw for 50% throttle ($I = P/V$)

Commented [SS11]: Todo: find power requirement for flying. How to monitor it

Commented [SS13]: Check correct value

Let's say we always fly with 80% rule – i.e, when battery is left with 20-30% we don't fly anymore.

- The payload impact the weight of the drone , so we have to increase the thriottle of motors. This will inturn impact the capacity , and how long drone can fly.
- Therefore we use our second option of 12000mAh, and do separate setup for LiDAR, Camera, Comm device , companion computer to understand who is power hungry and accordingly optimize the setup

Commented [SS14]: I don't think there is any better option than trial and run

5. Things to remember (possible mistakes as a beginner)

- Match the voltage with what brushless motors and Electronic Speed Controller (ESC) – regulate the speeds and signals the flight controller. Match the S-rating to motor and ESCs
- Ensure high C-Rating – for motor and payloads
- More weight to carry – need more power on motors so more capacity
- Be careful with connectors, adapters and soldering – done wrongly can increase resistance.
- Always fit the battery in the compartment before flying
- Flight time calculations pay attention – fly below the theoretical flight time (because wind, flight height, temperature – are some uncertainties)
- Charging, monitoring and Storage mode (min discharge 3.5V). Don't allow over discharging below 3.3 V
- Storage in fireproof bags. Don't leave battery unattended
- Have sand bags, or cement block near stored battery – in case of explosion.

Commented [SS15]: According to initial research with videos, forums, Chatgpt – 6S is suggested for Holybro X650
X650 uses individual ESCs
This draw upto 45A - so need battery above 3S.
Todos: Calculate actually how much power each motor draws.

Used resources:

1. <https://docs.px4.io/main/en/config/battery.html>
4. https://www.google.com/search?q=matek+sys+pdb+board&rlz=1C1GCEU_deDE1112DE1112&oq=matek+sys+pdb+board&gs_lcrp=EgZjaHJvbWUyBggAEEUYOTIICAFAQBqWGB4yDQgCEAAyhgMYgAQYigUyDQgDEAAyhgMYgAQYigUyDQgFEAAyhgMYgAQYigUyBwgFEAAy7wUyCggHEAAyAQYogQyCggIEAAyAQYogQyCggIEAAyAQYogTSAQC1MDNqMGo3qAIAA&sourceid=chrome&ie=UTF-8 (Battery breakdown – Energy density)
5. <https://luck.synhera.be/bitstream/handle/123456789/1760/Autopilot%20and%20companion%20computer%20for%20unmanned%20aerial%20vehicle%20Survey%20-%20Di%20Naro%20-%20Delvaux%20-%20HEPL.pdf?sequence=1&isAllowed=y>
6. <https://iotcart.in/product/matek-systems-pdb-xt60-power-distribution-board?srltid=AfmBOorFTFoF0fkPMFHvzqRhk3RQV5wkLhg9kZ2G5x7PzndY0Ot5gSCN>
7. <https://www.getfpv.com/electronics/power-distribution-boards-pdb.html?srltid=AfmBOoqVtUcL53EqCE8VhEJdqyIZRIjEmIplOmmn13dUqNCGAGTCIAA>
8. https://www.reddit.com/r/diydrones/comments/urp31c/is_a_power_distribution_board_a_must/
9. https://www.reddit.com/r/robotics/comments/10oj6c0/how_to_power_distribution_board_for_drones_need/
10. <https://dronenodes.com/pdb-power-distribution-board/#main>
11. <https://www.youtube.com/watch?v=MqEOXPdAJJU>

12. <https://www.youtube.com/watch?v=QhW-0ddGJ7A>
13. <https://www.youtube.com/watch?v=Fells7qG8JE>
14. <https://www.youtube.com/watch?v=nWOBa9TA2gY&list=PLwoDb7WF6c8kul5UdXNbB-Y9Kxhu0FAt3> (PID and filter ,Motor ESC protocols)